An Explainable Adaptive Temporal Learning Framework for Early Plant Growth Forecasting Using Integrated Soil Environmental and Historical Growth Data

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Abstract— Plant growth reflects interacting soil, environmental, and historical conditions. This paper presents an explainable adaptive temporal learning framework that forecasts future plant height using soil records, daily environment observations, and plant-specific growth history. It combines leakage-safe temporal features with persistence, linear, Random Forest, XGBoost, GRU, LSTM, and validation-trained fusion models. SHAP links explanations to the exact fitted artifact and input. Evaluation uses chronological splits and regression metrics. The project is implemented through React, Spring Boot, MySQL, and FastAPI. Since the supplied dataset has no repeated height target, the paper reports a reproducible design and does not fabricate performance claims.

Index Terms— Plant growth forecasting, temporal learning, explainable AI, SHAP, feature selection, precision agriculture.

I. INTRODUCTION

Plant growth monitoring supports data-driven cultivation, but current conditions alone do not represent the accumulated effects of prior growth, weather and soil. The proposed system forecasts a future plant-height value in centimetres at a declared horizon. Every completed prediction retains the origin timestamp, model version, feature pipeline version and input snapshot required for audit.

Conventional tabular learning represents nonlinear interactions but can treat observations independently. Plant development is temporal: earlier height and environmental exposure influence later growth. The framework therefore uses timestamp-aware joins, per-plant history, lagged values, rolling statistics and an explicit horizon definition. It is a research decision-support system; it does not diagnose disease, guarantee yield or treat SHAP attribution as causation.

A. Contributions

The contribution is a full-stack framework that integrates three modalities, compares tabular and temporal learners, performs adaptive feature selection using training data only, combines compatible branches through validation-trained fusion, and gives artifact-linked SHAP explanations.

II. RELATED WORK AND BASE PAPER CONTEXT

Recent work on greenhouse growth modelling demonstrates the value of environmental sensing, recurrent models and multimodal learning for plant-development trajectories. LSTM-based tomato-height studies show that historical conditions improve prediction relative to instantaneous inputs. Explainable AI is increasingly used to inspect feature contributions in agricultural models.

The selected base paper by Condò et al. introduces a self-adhesive fiber Bragg grating wearable that physically monitors stem elongation. It is a measurement system based on optical strain sensing. The present project is different: it forecasts a later numerical height from prior soil, environmental and growth records. The approaches can be complementary, but the proposed software does not reproduce or replace the sensor's physical measurement.

B. Research Gap

A practical gap remains for an auditable application that unifies soil provenance, daily environment records, repeated plant-height observations, temporal feature construction, adaptive selection, fusion and explanation under a chronological evaluation protocol.

III. PROPOSED FORECASTING FRAMEWORK

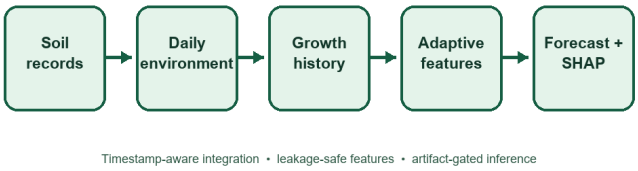


Fig. 1. Integrated plant-growth forecasting workflow.

At a forecast origin, records are aligned using plant identifier, location and real timestamps. Soil records are reused only when their temporal interpretation supports it. Missing values remain unknown; no missing height, moisture or weather value is invented. The feature pipeline uses only information available by the origin time.

A. Input Modalities

Modality	Examples
Soil	pH, moisture, nitrogen, phosphorus, potassium, texture
Environment	Temperature, humidity, precipitation, solar radiation
Growth	Plant ID, measurement timestamp, height, stage

Table I. Input modalities.

B. Base Paper and Project Comparison

Aspect	Base paper	This project
Goal	Measure elongation	Forecast future height
Input	FBG strain response	Soil, environment, history
Output	Physical measurement	Prediction + SHAP
Evaluation	Sensor metrology	Chronological MAE, RMSE, R squared

Table II. Base-paper distinction.

C. Target and Horizon

The primary research target is future plant height. A valid label is the real plant-height observation nearest to origin plus the configured horizon within a documented tolerance. Origins without a valid later measurement are excluded. Growth amount may be treated only as a separate labelled experiment, not mixed with the height target.

D. Temporal Feature Engineering

Examples include previous height, observed growth rate, lagged soil moisture where measured, rolling temperature and humidity summaries, cumulative precipitation, observation gaps and planting-age features. The pipeline prevents future observations from entering historical feature vectors.

IV. LEARNING MODELS AND FUSION

The persistence baseline predicts the latest measured height. Linear regression is included as a simple benchmark. Random Forest and XGBoost consume selected tabular features. GRU and LSTM consume ordered per-plant sequences. All comparable models use the same target definition and chronological partitions.

Model	Role
Persistence	Minimum future-height baseline
Linear regression	Linear tabular benchmark
Random Forest	Nonlinear tabular benchmark
XGBoost	Boosted tabular benchmark
GRU / LSTM	Temporal sequence benchmarks
Stacked fusion	Validation-trained integration

Table III. Model family.

A. Adaptive Feature Selection

Importance- or permutation-based selection is fitted on training observations only. The selected feature list, order, scaler and imputation settings are stored with the model artifact. Validation data may select a feature-selection configuration, but the held-out test set is never used to choose features.

B. Fusion Strategy

The primary fusion strategy is stacking. Base predictors are trained on the training period and generate temporally valid predictions. A meta-regressor is trained using those base outputs and selected context where justified. The configuration is selected on validation data, frozen, and then assessed once on the final test period.

C. SHAP Explainability

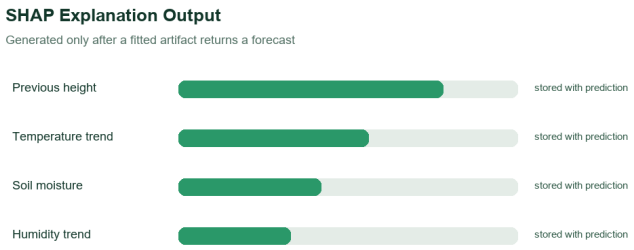


Fig. 2. Proposed SHAP output structure. Contributions are populated only for a fitted model and exact input.

Tree explainers are used for tree models. Neural explanations require a validated explainer and representative background data. For stacked fusion, the final explanation must identify meta-model behaviour; base-model values alone must not be misrepresented as a complete fusion explanation.

V. FULL STACK IMPLEMENTATION

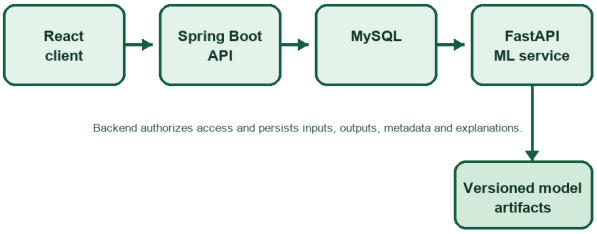


Fig. 3. React to Spring Boot to MySQL and FastAPI architecture.

React provides the user interface. Spring Boot authenticates users using JWT, enforces ownership, validates requests, stores observations and persists prediction metadata. MySQL retains plants, locations, soil/environmental/growth records, imports, predictions and evaluations. FastAPI owns preprocessing, artifact loading, forecasting, training and explainability.

A. Secure Prediction Workflow

The browser never calls the ML service directly. The backend retrieves only authorized records, builds a validated ML request, handles unavailable artifacts and timeouts, and saves the returned prediction with its model version and input snapshot.

B. Dashboard Output Design

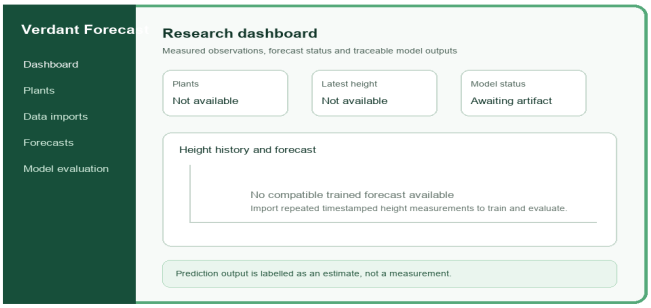


Fig. 4. Original project dashboard concept. It intentionally displays unavailable values until valid records and trained artifacts exist.

Measured values and forecasts are visually distinct. The result view labels output as “prediction, not measurement”, displays data-quality warnings, and later displays the measured future height and absolute error when it is recorded.

C. Data Quality

The system rejects invalid pH, impossible humidity, negative height, duplicate growth timestamps, unsupported horizons, unknown plants and insufficient history. It never silently changes missing values to zero. Import audit records preserve file name, row counts, rejection causes and validation warnings.

VI. EVALUATION AND RESULTS REPORTING

Forecasting experiments require chronological partitions rather than random mixing. The earliest period fits preprocessing and models; the following period selects hyperparameters and fusion; the newest period is untouched until final assessment. The paper must report date ranges, sample count and target/horizon alongside each metric.

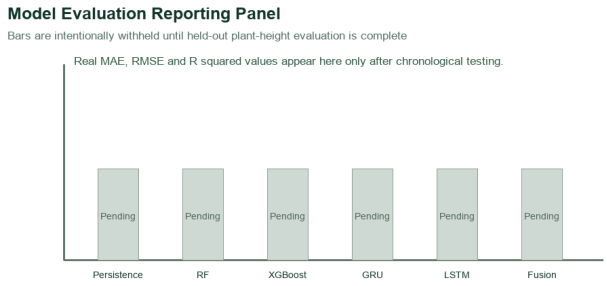


Fig. 5. Performance-reporting layout. Actual model bars are withheld until valid chronological evaluation is run.

A. Metrics

Metric	Meaning
MAE	Average absolute future-height error in cm
RMSE	Penalizes larger height errors
R squared	Explained variation; guarded for small/constant sets

Table IV. Regression metrics.

B. Current Dataset Observation

The supplied aggregate greenhouse file contains environmental and aggregate fruit/biomass fields but marks plant_height_cm as not provided. It cannot train or validate a plant-height forecaster. That constraint is recorded instead of hidden.

C. Results Status

Model	7-day result
Persistence	Not evaluated: height data required
Random Forest	Not evaluated: height data required
XGBoost	Not evaluated: height data required
GRU / LSTM	Not evaluated: sequences required
Fusion	Not evaluated: valid base artifacts required

Table V. Current evaluation status.

No numerical accuracy, MAE, RMSE, R squared, SHAP ranking, or “best model” claim is inserted before an experiment produces those results. This protects the integrity of the conference submission. Once repeated timestamped height measurements are available, the training pipeline produces the metrics table and real chart from persisted evaluation outputs.

VII. CONCLUSION AND FUTURE WORK

The proposed framework brings together authenticated observation management, leakage-safe temporal learning, adaptive selection, fusion, SHAP and chronological validation for early plant-growth forecasting. The next research step is collection or licensing of repeated per-plant height data at a frequency sufficient for the selected horizon. Future work includes uncertainty estimates, cross-location validation, species transfer and integration of real sensor measurements as additional inputs.

VIII. APPLICATION WORKFLOW

Step	Outcome
Register plant and location	Stable plant and location identifiers
Store observations	Source, timestamp and quality status
Request forecast	Validated artifact-gated inference
Save prediction	Input snapshot, version and SHAP output
Record actual measurement	Absolute error and evaluation history

Table VI. End-to-end workflow.

A. Reproducibility Checklist

Requirement	Implementation
Target	Future plant height in cm; declared horizon/tolerance
Split	Chronological train, validation and held-out test
Transforms	Fit on training period only
Artifacts	Store feature order, pipeline/model versions and configuration
Interpretation	SHAP linked to exact model and input; no causal claim

Table VII. Minimum reproducibility controls.

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